

INSTITUTIONAL COURSE SYLLABUS TEMPLATE

Course Code: STAT101	Course Title: Introduction to Probability theory and Statistics
Course Instructor: Dr.Sarah Aljohani	Email: sjohani@psu.edu.sa
Credit Hours: 3+1	Course Location: PSU
Scheduled Office Hours:	Sun, Mon, Tue 11-12 Wed:12-1, Thu:8-9.
Office Location:	R206
Co-Requisite:	Prerequisite: MATH002
On Campus or Online	On-Campus

Mission Statement

The Department of Mathematics and Sciences is committed to offering high-quality education in the mathematics and sciences field to enhance students' logical thinking and analytical skills. The department supports research in different scientific disciplines with distinct societal and scientific impacts.

I: Course Description:

Introduction to Statistics & Probability Theory is the first course in Statistics, designed to expose the students to the use of tables and graphs as a way to summarize data and provide concise and meaningful presentations for the data. Using the computing facilities at campus, problem-solving is spreadsheet-oriented whenever possible. Excel is a spreadsheet tool for manipulating quantitative data. The extent to which the variables are related is discussed deeply through graphs and using rules. The emphasis is on analyzing, understanding, and using statistics to solve real-world problems. Coverage of the course includes descriptive statistics, probability and probability distributions, and sampling.

II. Course Learning Outcomes: On the successful completion of this course, students will be able to demonstrate the following:

<u>Skills</u>	<u>Course Learning Outcomes</u>	<u>Measured by</u>
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<u>Knowledge and understanding</u>	1.1 CLO-1: Describe Data, Population, sample, parameters, statistic and variables. 1.2 CLO-2 Explain the concepts of descriptive statistics and probability rules.	Quizzes Major Exams Final Exam
<u>Skills</u>	2.1 CLO-3: Calculate the central tendencies, location, dispersion, Discrete and continuous probability distributions. 2.2 CLO-4: Illustrate the data through suitable statistics and graphical display. 2.3 CLO-5: Formulate a research study through survey and group projects using computer skills.	Quizzes Major Exams Final Exam
Values		

• **Tentative Weekly Course Schedule:**

Chapter	Topics	No. of Weeks	Contacts Hrs.
•	The Where, Why, and How of Data Collection	1	3 hours
•	Graphs, Charts, and Tables—Describing Your Data	2	8 hours
•	Describing Data Using Numerical Measures	2.5	10 hours
•	Introduction to Probability	2.5	10 hours

•	Discrete Probability Distributions	2	8 hours
•	Introduction to Continuous Probability Distributions	2	8 hours
•	Introduction to Sampling Distributions	2	8 hours
•	Estimating Single Population Parameters	1.75	5 hours

III. Tentative Weekly Course Schedule: (Should mention the specific course topics to be covered within the semester) *May change to accommodate guest presenters & student needs.*

WEEK	UNIT/ TOPIC	CLO(s) alignment	STUDENT T A S K S (assignments, r e a d i n g s , exams, quizzes, presentations, etc...)	Number of Contact hours
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1 Dec.29-02 Jan	1.1 and 1.2 Quickly, 1.3 Populations, Samples, and Sampling Techniques.	CLO:1	In-Class Exercises On-Line Homework Group discussions.	4
2 Jan 5-9	1.3 Populations, Samples, and Sampling Techniques. 1.4 Data Types and Data Measurement Levels. 2.1 Frequency Distributions and Histograms	CLO: 1,3,4	In-Class Exercises On-Line Homework Group discussions. Discussing different business applications	4
3 Jan 12-16	• <u>F</u> <u>re</u> <u>q</u> <u>u</u> <u>e</u> <u>n</u> <u>cy</u> <u>D</u> <u>is</u> <u>tr</u> <u>ib</u> <u>ut</u> <u>io</u> <u>n</u> <u>s</u> <u>a</u> <u>n</u> <u>d</u> <u>H</u> <u>is</u> <u>to</u> <u>gr</u> <u>a</u> <u>m</u> <u>st</u> • <u>B</u> <u>ar</u> <u>e</u> <u>h</u> <u>ar</u> <u>ts</u>	CLO: 3	Applying Statistical Tools using Excel Data Analysis Tool Pak In-Class Exercises On-Line	4

5 Jan 26-30	• <u>M</u> <u>e</u> <u>as</u> <u>u</u> <u>re</u> <u>s</u> <u>of</u> <u>C</u> <u>e</u> <u>nt</u> <u>a</u> <u>n</u> <u>d</u> <u>L</u> <u>e</u> <u>c</u> <u>at</u> <u>io</u> <u>n</u> 3.2 Measure of Variation	CLO:1,3	Applying Statistical Tools using Excel Data Analysis Tool Pak In-Class Exercises On-Line Homework	4
6 Feb. 2-6	3.3 Using the Mean and Standard Deviation Together. 4.1 The Basics of Probability	CLO:2,3	Applying Statistical Tools using Excel Data Analysis Tool Pak In-Class Exercises On-Line Homework Business applications and computer database exercises.	4
7 Feb. 9-13 Major-1 10th Feb Monday 12:00 to 1:00 Chapter 1, 2, 3	4.2 The Rules of Probability	CLO:1,2,3	In-Class Exercises On-Line Homework Business applications and computer data base exercises.	4

8 Feb 16-20	5.1 Introduction to Discrete Probability Distributions 5.2 The Binomial Probability Distribution	CLO:2,3	In-Class Exercises On-Line Homework Group discussions. Discussing different business applications	4
23 Feb.		Saudi Founding day		
9 Feb. 24-27	5.3 Other Discrete Probability Distributions.	CLO:2,3	In-Class Exercises On-Line Homework Group discussions. Discussing different business applications	4
28 th Feb. – 5 th April		Eid ul Fitar Holidays		
10 April 6-10 Quiz-2 Chapter 4 Monday 7 Apr	• <u>T</u> <u>h</u> <u>e</u> <u>N</u> <u>or</u> <u>m</u> <u>al</u> <u>p</u> <u>ro</u> <u>b</u> <u>a</u> <u>bi</u> <u>lit</u> <u>y</u> <u>D</u> <u>is</u> <u>tr</u> <u>ib</u> <u>ut</u> <u>io</u> <u>n</u>	CLO:2,3	In-Class Exercises On-Line Homework Group discussions. Discussing different business applications	4

11 April 13-17 Project	• <u>S</u> <u>a</u> <u>m</u> <u>pl</u> <u>in</u> <u>g</u> <u>E</u> <u>rr</u> <u>or</u> <u>:</u> <u>W</u> <u>h</u> <u>at</u> <u>it</u> <u>is</u> <u>a</u> <u>n</u> <u>d</u> <u>W</u> <u>h</u> <u>x</u> <u>it</u> <u>H</u> <u>a</u> <u>p</u> <u>p</u> <u>e</u> <u>n</u> <u>s</u> <u>7.2 Sampling Distribution of the Mean.</u>	CLO:2,5	In-Class Exercises On-Line Homework G r o u p discussions. D i s c u s s i n g different business applications	4
12 April 20-24	7.3 Sampling Distribution of a Proportion.	CLO:4	In-Class Exercises On-Line Homework G r o u p discussions. D i s c u s s i n g different business applications	4

13 April 27- 1 May Major-2 28 April Monday 12:00 to 1:00 Chapter 4,5,6	• <u>P</u> <u>o</u> <u>n</u> <u>t</u> <u>a</u> <u>n</u> <u>d</u> <u>C</u> <u>e</u> <u>n</u> <u>s</u> <u>i</u> <u>d</u> <u>e</u> <u>n</u> <u>c</u> <u>e</u> Interval Estimates for A Population Mean	CLO:3,5	In-Class Exercises On-Line Homework G r o u p discussions. D i s c u s s i n g different business applications	4
14 May 6-8	<u>8.2</u> <u>Determining</u> <u>the Required</u> <u>sample size</u> <u>for</u> <u>Estimating a</u> <u>population</u> <u>mean</u>	CLO:2,4	In-Class Exercises On-Line Homework G r o u p discussions. D i s c u s s i n g different business applications	4
15 May 11-15	Revision	CLO:5	In-Class Exercises On-Line Homework G r o u p discussions. Discussing different business applications	4
Exams				

IV. Student Assessment & Teaching Strategies:

Domain	Assessment Task	Teaching Strategy
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<u>Knowledge and Understanding</u> <u>CLO-1: Describe Data, Population, sample, parameters, statistic and variables.</u> CLO-2 Explain the concepts of descriptive statistics and probability rules.	<u>Direct Assessment Tool</u> <u>Quiz, Majors</u> <u>Assignments</u> <u>Indirect</u> <u>Assessment Tool:</u> <u>Course Exit Survey.</u>	<u>Lectures;</u> <u>Small group discussion;</u> <u>Problem Solving</u> <u>sessions;</u> Applying Statistical Tools using Excel Data Analysis Tool Pak
<u>Skills</u> CLO-3: Calculate the central tendencies, location, dispersion, Discrete and continuous probability distributions. CLO-4: Illustrate the data through suitable statistics and graphical display. CLO-5: Formulate a research study through survey and group projects using computer skills.	<u>Direct Assessment Tool</u> <u>Quiz, Majors</u> <u>Assignments</u> <u>Indirect</u> <u>Assessment Tool:</u> <u>Course Exit Survey.</u>	<u>Lectures;</u> <u>Small group discussion;</u> <u>Problem Solving</u> <u>sessions;</u> <u>Hands-on Practice;</u> <u>Examples presented in class;</u> Applying Statistical Tools using Excel Data Analysis Tool Pak
Values .		

<u>Project Rubric;</u>					
	1	0.75	0.5	0.25	Score(1)
Submission of project.	In Time	Late by one day	Late by two days	Late by three days	
	4	3	1	Score (4)	
Content, Accuracy & originality.	Excellent	Very Good	Satisfactory		
	4	3	1	Score (4)	
Using Excel to carry computation, at least two different type of charts and graphs	Excellent	Very Good	Satisfactory		
	1	0.75	0.5	Score(1)	
Submission of the project in APA style	Excellent	Very Good	Satisfactory		

V. Course Requirements Assessment for this course is based on 2 Quizzes, participation in the classroom, online homework, One Project, 2 Majors and final exam with the following percentage.

VI. Schedule of Assessment

<u>#</u>	<u>*Assessment task</u>	<u>Week Due</u>	<u>Proportion of Final Assessment</u>
<u>1</u>	<u>First Major Exam</u>	<u>Week 7, 10th Feb</u> <u>Monday</u>	<u>20%</u>

<u>2</u>	<u>Second Major Exam</u>	Week 13, 28 April Monday	<u>20%</u>
<u>3</u>	<u>Quiz 1</u>	<u>Week 4, 20th Jan</u> <u>Monday</u>	<u>4%</u>
<u>4</u>	<u>Quiz 2</u>	Week 10, 7th Apr Monday 7	<u>4%</u>
<u>4</u>	<u>Project (Turnitin</u> <u>through Moodle)</u>	<u>Week 11</u>	<u>6%</u>
<u>5</u>	<u>Online Homework+</u> <u>Assignments/</u> <u>Participation</u>	<u>Throughout the term</u>	<u>6%</u>
<u>6</u>	<u>Final Exam</u>		<u>40%</u>
<u>7</u>	<u>Total</u>		<u>100%</u>

VII. Learning Resources

- References –

<u>Required Textbooks</u>	Business Statistics by Groebner, Shannon, Fry. Tenth Edition
<u>Essential References Materials</u>	Statistics for Business by Farouk Benghezi. Arab World Edition
<u>Electronic Materials</u>	<u>Access to university LMS. We use the LMS to</u> <u>communicate with the students; announcements</u> <u>and grading are all conveyed to the students</u> <u>through this system.</u> <u>Computer Lab</u> <u>MyLab Math by Pearson to carry online home</u> <u>works.</u>
<u>Other Learning Materials</u>	<u>None</u>

- Facilities Required – (online – laptop/internet, lecture room, computing resources, laboratories, others)
- Digital Tools – The platform will be used for all online education experiences at PSU. It supports a variety of methods to support teaching and learning strategies across all disciplines. It also includes the LMS.
 - Learning Management System – It is required to make use of MOODLE. I will upload the syllabus and other course materials for your use and certain assignments.
 - Pearson – it is required to conduct homework.

- Learning Management System – LMS using “Moodle” is required; items such as syllabus, course materials, assignments, rubrics, and announcements will be delivered through Moodle.
- Online Courses: Google Meet – Not applicable.
- Writing and Tutoring Center – Students are highly advised to use the Writing and Tutoring Center’s academic services by booking tutoring/writing appointments through the booking system: (<https://psu.mywconline.net/>)

VIII. Classroom Policies

- Academic Integrity Policy (e.g., plagiarism or dishonesty)

Cheating, attempting to cheat, plagiarism, or violating instructions and examination regulations shall render the offender subject to punishment per the student disciplinary rules issued by the university council.

“Plagiarism can be defined as unintentionally or deliberately using another person’s writing or ideas as though they are one’s own. Plagiarism includes but is not limited to, copying another individual’s work and taking credit for it, paraphrasing information from a source without proper documentation, and mixing one’s own words with those of another author without attribution. In addition, buying a paper or project, or downloading a paper from the Internet, and submitting them as your own is also plagiarism. The penalty for academic dishonesty will bring course expulsion, failure, or even suspension” (Academic Integrity and Syllabus Acknowledgement Form).
All students are expected to submit their ‘own’ work and not the work of others.

- Attendance Policy

Please adhere to the following guidelines:

- The University attendance policy will be strictly followed. In this course, the absence of **(16)** hours results in a Denied Notice (DN). (Refer to the student regulation handbook for further details.)

Please Note: IT IS THE SOLE RESPONSIBILITY OF THE STUDENT TO KEEP SATISFYING their level of attendance. Otherwise, a “DN” grade will be granted automatically during any time in the semester without any notice from the e-register system.

- No make-up exams will be allowed except for university accepted documents. (Refer to the student regulation handbook for further details.)

- **Homework Submission Policy**

Online homework should be completed before the due date.